
```
% Kyle Kasmiersky
% Professor Luo - structures 3
% Homework 3
```

```
%-----
clc; clear; close all
```

Problem 5e

```
t = 0:0.001:40;
dt = 0.001;

x0=0;
v0=0;

x = zeros(1,length(t));
v = zeros(1,length(t));

const = -4/0.5;

for i = 2:length(t)
    x(i) = x(i-1) + dt.*(v(i-1));
    v(i) = v(i-1) + dt.*(const.*x(i-1) - 9.81);
end
```

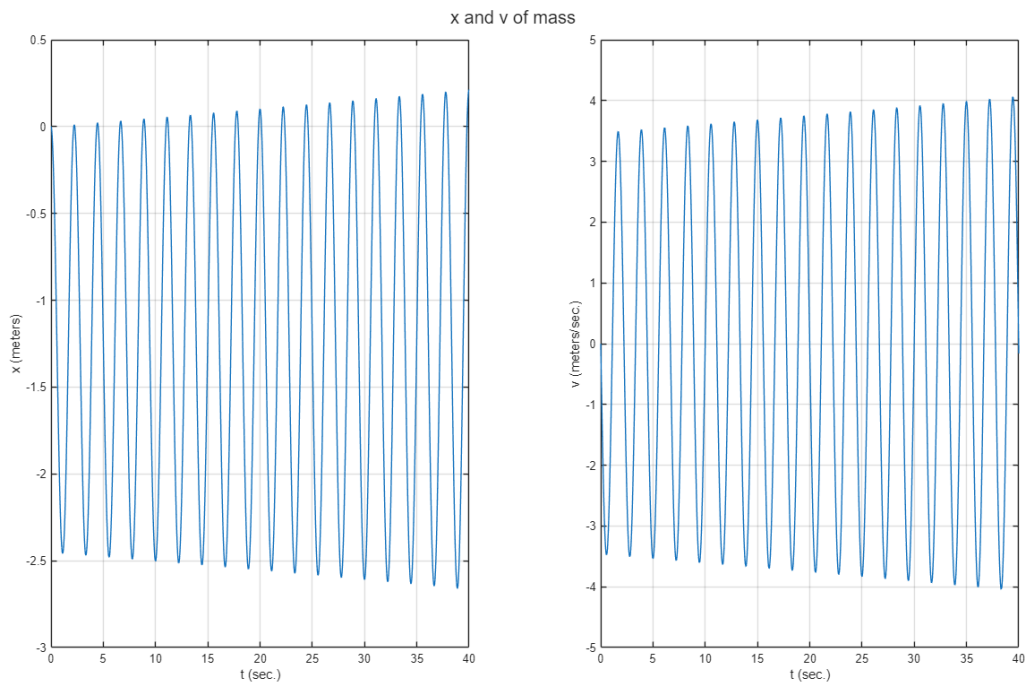
```
figure()
```

```
sgtitle("x and v of mass")
```

```
subplot(1,2,1)
plot(t,x)
xlabel("t (sec.)")
ylabel("x (meters)")
grid on
```

```
subplot(1,2,2)
plot(t,v)
xlabel("t (sec.)")
ylabel("v (meters/sec.)")
grid on
```

```
clear
```



Problem 5f

```
t = 0:0.001:40;
dt = 0.001;

x0=0;
v0=0;

x = zeros(1,length(t));
v = zeros(1,length(t));

const = -4/0.5;

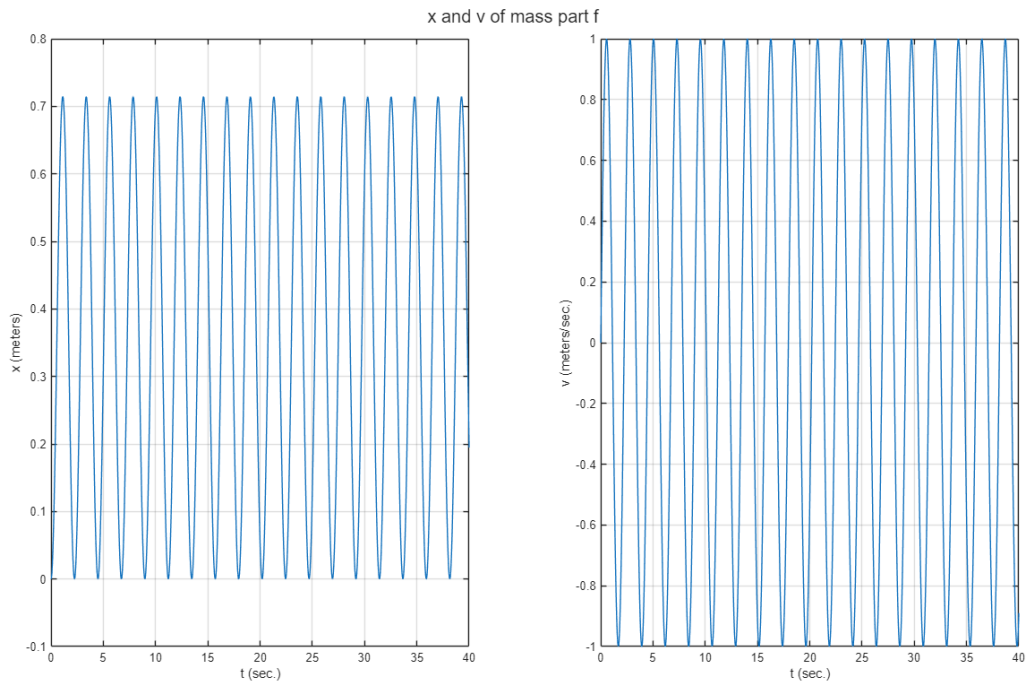
for i = 2:length(t)
    x(i) = x(i-1) + dt.*(v(i-1));
    v(i) = sin(2.8.*t(i));
end

figure()

sgtitle("x and v of mass part f")

subplot(1,2,1)
plot(t,x)
xlabel("t (sec.)")
ylabel("x (meters)")
grid on
```

```
subplot(1,2,2)
plot(t,v)
xlabel("t (sec.)")
ylabel("v (meters/sec.)")
grid on
```



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